Light

GCSE Computer Science — Legal, Ethical & Environmental Issues (Set 3)

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GCSE • Edexcel • Grades 3—9 | Topic test | 30 min | 55 marks | Answered: 0 / 30

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Multiple-choice topic test for **GCSE Computer Science (Edexcel)**. You may skip questions. When you submit, you'll see your score, a grade estimate, and worked explanations. **Tip:** use the stopwatch above to time yourself.

Q1 (1 mark)

Before installing software, users must typically agree to a legal document that specifies what they can and cannot do with the software.
What is this document called?

- ○ **A.** An End User Licence Agreement (EULA)
- ○ **B.** A driving licence
- ○ **C.** A passport application
- ○ **D.** A birth certificate

Q2 (1 mark)

A company's logo and brand name are protected so that competitors cannot use similar designs to confuse customers.
What type of intellectual property protection is this?

- ○ **A.** A patent
- ○ **B.** A trademark
- ○ **C.** A copyright
- ○ **D.** A trade secret

Q3 (1 mark)

Websites and apps that are likely to be used by children must follow additional rules about collecting and using their personal data.
What UK code provides guidance on this?

- ○ **A.** The Highway Code
- ○ **B.** The Building Code
- ○ **C.** The Age Appropriate Design Code (Children's Code)
- ○ **D.** The Dress Code

Q4 (1 mark)

Smart home devices like voice assistants, cameras, and thermostats collect data about residents' activities and habits.
What privacy concern does this raise?

- ○ **A.** The devices are too expensive
- ○ **B.** The devices use too much shelf space
- ○ **C.** The devices are too difficult to clean

○ **D.** Continuous monitoring creates detailed profiles of daily life that could be accessed by companies, hackers, or authorities

Q5 (1 mark)

A company discovers hackers have stolen customer data including names, emails, and passwords.
What must the company do under UK data protection law?

○ **A.** Report the breach to the ICO within 72 hours if it risks people's rights and freedoms

○ **B.** Keep the breach secret to avoid bad publicity

○ **C.** Wait at least one year before telling anyone

○ **D.** Only report breaches involving more than one million people

Q6 (1 mark)

During the pandemic, many office workers began working from home instead of commuting to offices.
What environmental impact did this have?

○ **A.** Reduced commuting emissions but increased home energy use; overall impact depends on commute distances, home heating efficiency, and whether offices closed or continued operating

○ **B.** Working from home has no environmental impact whatsoever

○ **C.** Commuting always has zero carbon emissions

○ **D.** Home heating uses no energy

Q7 (1 mark)

Social media and search engines show users content similar to what they have previously engaged with, potentially limiting exposure to different viewpoints.
What is this effect called?

- ○ **A.** Data compression
- ○ **B.** File encryption
- ○ **C.** A filter bubble or echo chamber
- ○ **D.** Network latency

Q8 (1 mark)

Video games sell randomised virtual items where players pay real money but do not know what they will receive.
Why do some argue these should be regulated like gambling?

- ○ **A.** Because the games are too colourful
- ○ **B.** Because the variable rewards use gambling-like mechanics and can encourage compulsive spending, especially in children
- ○ **C.** Because the games load too slowly
- ○ **D.** Because the games use too much storage

Q9 (1 mark)

Data derived from physical characteristics like fingerprints, facial geometry, or iris patterns is used to identify individuals.
What is this type of data called?

- ○ **A.** Biometric data

- B. Geographic data
- C. Financial data
- D. Calendar data

Q10 (1 mark)

Manufacturing a single smartphone requires mining materials from multiple continents, assembly in factories, and shipping worldwide. What is the term for all emissions generated throughout a product's creation and delivery?

- A. Operational carbon
- B. Direct emissions
- C. Embodied carbon or supply chain emissions
- D. Renewable energy

Q11 (1 mark)

When a person dies, their family may want access to their email, social media, and online accounts.
What complicates this 'digital inheritance'?

- A. Email servers are too loud
- B. The internet stops working when someone dies
- C. Online accounts automatically delete themselves
- D. Accounts are protected by passwords and terms of service that may not permit transfer to others, even family members

Q12 (1 mark)

Health organisations warn about excessive screen time, particularly for children, citing concerns about physical inactivity, sleep disruption, and impact on development.
What type of ethical concern is this?

- ○ **A.** A network security concern
- ○ **B.** A database design concern
- ○ **C.** A digital wellbeing concern about technology's impact on physical and mental health
- ○ **D.** A software compatibility concern

Q13 (2 marks)

A gym app requests access to users' contacts, photos, location, and microphone, even though it only needs to track workouts.
Which data protection principle does this potentially violate?

- ○ **A.** Data minimisation — organisations should only collect personal data that is necessary for their stated purpose; requesting excessive permissions violates this principle
- ○ **B.** Data maximisation — organisations should collect as much data as possible
- ○ **C.** Data randomisation — all data should be collected randomly
- ○ **D.** Data acceleration — data must be collected as quickly as possible

Q14 (2 marks)

A person gains unauthorised access to a computer system with the intention of using the access to commit blackmail.

Which section of the Computer Misuse Act 1990 applies?

- ○ **A.** No section applies because blackmail is a separate crime
- ○ **B.** Section 2 — unauthorised access with intent to commit or facilitate further offences, which covers access intended to enable other crimes like blackmail
- ○ **C.** Only the Copyright Act applies
- ○ **D.** The Data Protection Act applies instead

Q15 (2 marks)

Police use algorithms that analyse historical crime data to predict where crimes are likely to occur and deploy officers accordingly.
What ethical concerns does predictive policing raise?

- ○ **A.** The algorithms use too much electricity
- ○ **B.** Police officers prefer paper maps
- ○ **C.** Algorithms cannot process geographic data
- ○ **D.** Historical data reflects past policing patterns, not just crime; areas heavily policed show more recorded crime, creating feedback loops; this can perpetuate discrimination against communities already over-policed

Q16 (2 marks)

A small number of large technology companies dominate search, social media, mobile operating systems, and cloud computing.
What concerns does this market concentration raise?

- ○ **A.** Large companies have better office furniture

○ **B.** Market dominance can reduce competition and innovation, give companies excessive power over users and smaller businesses, enable data accumulation that reinforces advantages, and raise concerns about influence over public discourse

○ **C.** Small companies have faster websites

○ **D.** Market concentration only affects company shareholders

Q17 (2 marks)

Employees argue they should not be expected to respond to work emails or messages outside working hours, even though technology makes them constantly reachable.
What is this proposed right called?

○ **A.** The right to faster internet

○ **B.** The right to free software

○ **C.** The right to disconnect — establishing boundaries between work and personal time, protecting mental health and work-life balance from technology-enabled always-on work culture

○ **D.** The right to larger monitors

Q18 (2 marks)

A phone manufacturer stops providing security updates for devices after three years, making them vulnerable to attacks even if hardware still works perfectly.
What is this practice and its environmental impact?

○ **A.** Software obsolescence that forces early replacement of functional hardware, generating unnecessary e-waste and wasting the embodied carbon from manufacturing

- ○ **B.** Security update support — it extends device lifespan
- ○ **C.** Hardware improvement — older devices become faster
- ○ **D.** Energy efficiency — older devices use less power

Q19 (2 marks)

Apps continuously track users' locations, building detailed histories of everywhere they go.
Why is continuous location tracking particularly sensitive?

- ○ **A.** Location data uses too much storage space
- ○ **B.** Location history reveals sensitive information — home address, workplace, religious attendance, medical visits, political activities, relationships — that users may not realise they are sharing
- ○ **C.** Location tracking makes phones heavier
- ○ **D.** Location data is always completely anonymous

Q20 (2 marks)

Users encounter so many cookie banners, permission requests, and consent forms that they click 'accept' without reading, just to access content.
What problem does this illustrate?

- ○ **A.** Cookie banners are too colourful
- ○ **B.** Websites have too many pictures
- ○ **C.** Consent fatigue — overwhelming users with consent requests degrades the quality of consent, as users mechanically agree without understanding; this undermines the purpose of consent requirements

- ○ **D.** Internet connections are too fast

Q21 (2 marks)

Schools use AI systems that monitor student behaviour, predict academic performance, and personalise learning.
What ethical considerations should guide AI use in education?

- ○ **A.** AI systems should monitor students as intensively as possible
- ○ **B.** Schools should avoid all technology use entirely
- ○ **C.** Only AI companies should decide how AI is used in schools
- ○ **D.** Student privacy (especially for minors), accuracy and bias in predictions, effects on student autonomy, data retention limits, transparency about how AI affects opportunities, and ensuring AI supports rather than replaces teacher judgment

Q22 (2 marks)

Some argue everyone should be required to use their real identity online, while others defend anonymous speech.
What are the key considerations in this debate?

- ○ **A.** Real names make websites load faster
- ○ **B.** Anonymous speech enables harassment but also protects whistleblowers, abuse victims, political dissidents, and those exploring sensitive topics; real name requirements may reduce some harms while creating others
- ○ **C.** Anonymity only benefits criminals
- ○ **D.** Everyone already uses their real name online

Q23 (2 marks)

Cities deploy sensors throughout public spaces to monitor traffic, air quality, noise levels, and pedestrian flows to improve urban management.
What trade-offs do these 'smart city' technologies involve?

- ○ **A.** Sensors improve traffic and environmental management but also enable pervasive public surveillance; benefits must be weighed against privacy implications, data security risks, and questions about consent in public spaces
- ○ **B.** Smart city sensors have no privacy implications
- ○ **C.** Sensors only collect data about weather
- ○ **D.** All sensor data is automatically deleted immediately

Q24 (3 marks)

A technology company proposes using anonymised NHS health records to train AI systems that could improve disease diagnosis.
What ethical considerations should inform decisions about using health data for AI development?

- ○ **A.** Commercial entities should have unrestricted access to all health data
- ○ **B.** Health data should never be used for any research purposes
- ○ **C.** All health data decisions should be made without patient involvement
- ○ **D.** Genuine anonymisation difficulty (re-identification risks), patient consent and expectations (NHS data given for treatment, not commercial AI), public vs private benefit balance, governance ensuring appropriate use, transparency about data flows, and whether patients should share in commercial value derived from their data

Q25 (3 marks)

A person registers domain names containing famous brand names (like 'nike-shoes-official.com') hoping to sell them to the trademark owners at inflated prices.
What legal and ethical issues does this raise?

- ○ **A.** This is legitimate domain investment with no legal issues
- ○ **B.** Cybersquatting (registering domains in bad faith to profit from others' trademarks) is both unethical and illegal; dispute resolution processes (UDRP) and trademark law enable brand owners to recover domains; it wastes resources, enables fraud, and exploits first-come-first-served domain systems
- ○ **C.** Only the domain registrar can decide who owns domains
- ○ **D.** Trademark law does not apply to domain names

Q26 (3 marks)

A hospital's computer systems are encrypted by ransomware. Attackers demand payment to restore access to patient records needed for treatment.
What ethical dilemmas does this situation present?

- ○ **A.** Hospitals should always pay ransoms immediately without consideration
- ○ **B.** Ransomware attacks on hospitals are not serious because backups exist
- ○ **C.** The situation only affects IT departments, not patient care
- ○ **D.** Paying may save patients now but funds criminal enterprises and encourages future attacks on healthcare; not paying upholds principles but risks patient harm; decisions involve immediate patient welfare vs systemic effects; prevention investment and backup systems reduce future dilemmas

Q27 (3 marks)

A country requires that all data about its citizens must be stored on servers physically located within its borders, not in foreign data centres. What are the arguments for and against such 'data sovereignty' requirements?

- ○ **A.** Data localisation has no impact on anything
- ○ **B.** All data should always be stored in a single global location
- ○ **C.** Only technology companies should decide where data is stored
- ○ **D.** Arguments for: protection from foreign government access, supporting local technology industry, ensuring local law applies, national security concerns. Arguments against: increased costs and reduced efficiency, potential for authoritarian control, conflicting requirements creating internet fragmentation, and reduced access to global services

Q28 (3 marks)

AI and automation are advancing rapidly. Workers in many fields face the possibility that their skills will become obsolete faster than they can retrain.
How should society address technology-driven skill obsolescence?

- ○ **A.** Workers alone are responsible for predicting and adapting to technological change
- ○ **B.** All automation should be banned to preserve existing jobs
- ○ **C.** Skill obsolescence is not a real concern because technology creates equal replacement jobs
- ○ **D.** Shared responsibility: employers should invest in workforce transition; government should fund retraining and provide safety nets; education should focus on adaptable skills; technology deployment

pace may need consideration; workers cannot solely bear adaptation costs of changes that benefit society broadly

Q29 (3 marks)

Emerging neurotechnology can read brain signals for device control, potentially detect thoughts, emotions, and mental states.
What new categories of rights might need protection as this technology develops?

- ○ **A.** Neurotechnology has no implications for rights because it only reads physical signals
- ○ **B.** Cognitive liberty (freedom from interference with mental processes), mental privacy (protection of thoughts from involuntary access), mental integrity (protection from manipulation), and psychological continuity (right to maintain personal identity) — collectively called 'neurorights' — may need explicit protection as mind-reading technology advances
- ○ **C.** Existing privacy laws fully address all neurotechnology concerns
- ○ **D.** Only technology companies should define neurorights

Q30 (3 marks)

Thousands of satellites are being launched for internet connectivity and other purposes, creating space debris that may threaten future space access.
What environmental and governance issues does the growth of satellite constellations raise?

- ○ **A.** Space activities have no environmental considerations
- ○ **B.** Only national space agencies launch satellites
- ○ **C.** Debris accumulation may create cascading collisions (Kessler syndrome) making orbits unusable; light pollution affects astronomy; lack of international governance allows unilateral actions affecting

shared space; end-of-life disposal responsibilities are unclear; equitable access concerns arise as limited orbital slots are claimed

○ **D.** Satellite numbers are strictly limited by international agreement

Total: **55** marks • Time: **30 min**

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